From the energy envelope to a two-chirality topological gap

23 September 2026. This note composes the existing functional-pencil theorem with both Chern–Weil identities. Its new application is a uniform integral Einstein obstruction on a realized family of mixed blow-ups and handle additions lying away from the previously recorded one-third critical band. All statements depend on the analytic theorem in the functional-pencil manuscript, through the energy-envelope theorem. The elementary Chern–Weil calculation is classical. No independent literature-priority certification is claimed for this composition.

1. An upper bound that applies to every candidate

Let

$$M = X\#k\overline{\mathbb{CP}}^2\#m(S^1\times S^3), \qquad A = c_1^2(X) > 0, \quad h = \frac{k+4m}{A} \in (0, 1/3), \tag{1}$$

where X is a minimal complex surface of general type. Write

$$T_\pm = 2\chi(M) \pm 3\tau(M), \qquad U = \frac{3\min(T_+, T_-)}{A} = \frac{3(2\chi(M) - 3|\tau(M)|)}{A},$$

$$R = \int_M |\operatorname{Ric}|^2, \quad P = \int_M |\operatorname{Ric}^\circ|^2, \quad e = \frac{R}{8\pi^2 A}, \quad F(h) = \frac{(4-\sqrt{1-3h})^2}{9}. \tag{2}$$

The Ricci norms are tensor norms; Weyl norms are Hilbert–Schmidt endomorphism norms on two-forms. Set

$$\mathscr{E}(g) = \inf_{\lambda\neq 0} \frac{\int |\operatorname{Ric} - \lambda g|^2}{\lambda^2 \operatorname{Vol}(g)}. \tag{3}$$

The scalar monopole estimate makes $R > 0$. Direct minimization, with $I = \int s$ and $V = \operatorname{Vol}(g)$, gives

$$\mathscr{E} = 4 - \frac{I^2}{RV} \geqslant \frac{4P}{R}. \tag{4}$$

For $I = 0$, the value four is an infimum as $|\lambda| \to \infty$; the same formula remains valid. The inequality uses $I^2/V \leqslant \int s^2 = 4(R - P)$, not a sign assumption on s.

The two Chern–Weil formulas give, separately for either sign,

$$\int |W^\pm|^2 = 2\pi^2 T_\pm - \frac{R}{12} + \frac{P}{3} \geqslant 0.$$

Consequently

$$e \leqslant U + \frac{4P}{8\pi^2 A} \leqslant U + e\mathscr{E}, \qquad \boxed{e(1-\mathscr{E}) \leqslant U.} \tag{5}$$

For Einstein metrics, this is the genuine upper bound $e \leqslant U$. It controls all candidates, regardless of their proximity to a preferred construction or variational minimum. No energy infimum has been used as an upper bound. No monopole structure in the opposite orientation is assumed: only the nonnegativity of its Weyl energy is used.

2. A uniform topological inequality for all smooth metrics

Theorem 1. There is a universal $c > 0$ such that every metric on every manifold (1) satisfies

$$\boxed{\mathscr{E}(g) + (U - F(h))_+ \geqslant ch^2.} \tag{6}$$

One may take $c = \varepsilon/7$, where $0 < \varepsilon \leqslant 1$ is the constant in the energy-envelope theorem

$$\mathscr{E}(g) + (e - F(h))_+ \geqslant \varepsilon h^2. \tag{7}$$

The same argument works in the abstract actual-class setting of that theorem: $T_+ = A(1-h)$, and for every metric an actual conformally solvable Spin^c structure has $(a_g^+)^2 \geqslant A$ and $a_g^2 \geqslant T_+$.

Proof. Since $T_+/A = 1-h$, one has $U \leqslant 3$. If $\mathscr{E} \geqslant 1/2$, (6) is immediate because $ch^2 \leqslant 1/63$. If $\mathscr{E} < 1/2$, (5) forces $U > 0$ and

$$e \leqslant \frac{U}{1-\mathscr{E}} \leqslant U + 2U\mathscr{E} \leqslant U + 6\mathscr{E}.$$

Hence

$$(e - F(h))_+ \leqslant (U - F(h))_+ + 6\mathscr{E}.$$

Substitute this into (7) and use $7\mathscr{E} + (U-F)_+ \leqslant 7[\mathscr{E} + (U-F)_+]$. This proves (6), including $U \leqslant 0$ through the first case. □

Corollary 2. If

$$U \leqslant F(h) + \frac{c}{2}h^2, \tag{8}$$

then every smooth metric satisfies

$$\mathscr{E}(g) \geqslant \frac{c}{2}h^2. \tag{9}$$

There are no Einstein metrics of any sign on this manifold. Nonzero Einstein constants would give $\mathscr{E} = 0$. For the zero constant, the signed scalar monopole inequality $\int s_-^2 \geqslant 32\pi^2 A > 0$ excludes Ricci-flatness separately. For either normalized pointwise branch,

$$\delta_\sigma(g) = \|g^{-1}\,\mathrm{Ric} - \sigma I\|_{\infty,\mathrm{op}}, \qquad \sigma \in \{-1, 1\},$$

one has $\mathscr{E} \leqslant 4\delta_\sigma^2$. Thus

$$\boxed{\delta_\sigma(g) \geqslant h\sqrt{c/8}.} \tag{10}$$

This is an obstruction to both pointwise inputs on these types, not a Yau26 counterexample.

The previously obtained family-uniform theorem used only $T_+/A = 1 - h$. Formula (6) adds the independent opposite-chirality budget T_-. When T_- is large it yields no new restriction; the mixed examples below have a small enough T_- for it to matter.

3. Realized mixed ball-quotient examples outside the old band

Choose any compact smooth complex-hyperbolic surface X_0, with complex orientation, and a sequence of connected finite unramified covers $X_j \to X_0$ of unbounded degree. Such a sequence exists: the cocompact lattice is an infinite finitely generated linear group in characteristic zero (use its faithful adjoint representation), hence is residually finite. A finitely generated residually finite infinite group has finite-index subgroups of unbounded index. Otherwise it has only finitely many

finite-index subgroups, their intersection is finite-index, and residual finiteness makes that intersection trivial, forcing the group to be finite. All covers remain compact complex hyperbolic and minimal of general type.

Put $A_j = c_1^2(X_j) \to \infty$. The Chern numbers are

$$\chi(X_j) = A_j/3, \qquad \tau(X_j) = A_j/9.$$

Define integers

$$k_j = \left\lfloor \frac{29A_j}{2025} \right\rfloor, \qquad m_j = \left\lfloor \frac{397A_j}{32400} \right\rfloor - 2, \qquad M_j = X_j \# k_j \overline{\mathbb{CP}}^2 \# m_j(S^1 \times S^3). \tag{11}$$

For sufficiently large j, both integers are positive. No prescribed divisibility of the cover degree or A_j is needed.

Let

$$h_0 = \frac{19}{300}, \quad x_0 = \frac{29}{2025}, \quad z_0 = \frac{397}{32400}, \quad F_0 = F(h_0) = \frac{961}{900}.$$

Then $x_0 + 4z_0 = h_0$, and

$$1 + 15x_0 - 12z_0 = F_0. \tag{12}$$

Write $k_j = x_0 A_j - \alpha_j$ and $m_j = z_0 A_j - \beta_j - 2$, where $0 \leqslant \alpha_j, \beta_j < 1$. Consequently

$$h_j = h_0 - \frac{\alpha_j + 4\beta_j + 8}{A_j},$$

$$\frac{3T_-(M_j)}{A_j} = 1 + \frac{15k_j - 12m_j}{A_j} = F_0 + \frac{-15\alpha_j + 12\beta_j + 24}{A_j}. \tag{13}$$

The signature is positive, since $\tau(M_j)/A_j = 1/9 - k_j/A_j \geqslant 1/9 - x_0 > 0$. Thus the second expression in (13) is exactly U_j. Moreover

$$F'(h) = \frac{4 - \sqrt{1-3h}}{3\sqrt{1-3h}}, \qquad 0 < F'(h) \leqslant \frac{31}{27} < 2 \quad (0 \leqslant h \leqslant h_0).$$

For large j, therefore,

$$0 < h_j < h_0, \qquad \boxed{0 < U_j - F(h_j) < \frac{62}{A_j}.} \tag{14}$$

Indeed, the second correction in (13) lies strictly between $9/A_j$ and $36/A_j$, and $0 < F_0 - F(h_j) < 26/A_j$.

Since $h_j \to h_0 > 0$, (14) eventually implies (8). We have proved:

Theorem 3. The realized sequence (11), for every sufficiently large j, satisfies

$$\boxed{\inf_{g \in \mathrm{Met}(M_j)} \mathscr{E}(g) \geqslant \frac{ch_0^2}{8} > 0.} \tag{15}$$

In particular, none of these smooth manifolds carries any Einstein metric. For every smooth metric on every sufficiently large member,

$$\delta_\sigma(g) \geqslant h_0\sqrt{c/32}, \qquad \sigma = \pm 1. \tag{16}$$

The threshold index depends on the universal analytic constant and the chosen cover sequence; no explicit numerical lower bound for that constant is claimed. The manifolds have unbounded Euler characteristic, so an infinite subsequence consists of pairwise distinct smooth types.

Actual gauge input, not merely numerical Chern data

For every metric on (11), select the exceptional signs at its polarization as in Section 2 of the polarized closure proof. The actual structure has

$$(a_g^+)^2 \geqslant A_j, \qquad a_g^2 = A_j - k_j \geqslant A_j - k_j - 4m_j = T_+(M_j).$$

Nonvanishing after handle addition includes the m_j loop insertions. The structure is fixed throughout the conformal class; the unchanged Hodge star on two-forms supplies the same class for every auxiliary positive weight. This is precisely the quantifier $\forall g\, \exists \mathfrak{s}_g\, \forall f > 0$ used in the energy-envelope theorem. The classical input is LeBrun's Proposition 3.2 and its polarized proof, not residual finiteness or the intersection form alone. Alternatively, sufficiently high covers have $b_+ > 1$, since $\chi(\mathscr{O}_{X_j}) = A_j/9 \to \infty$ and $b_+ = 1 + 2p_g$; hence these examples do not require the small-b_+ chamber extension.

4. Precisely which prior tests these examples pass

The following comparisons establish additional coverage relative to the recorded first-line bounds and to the displayed classical tests. They do not prove that every Einstein-obstruction paper has been exhaustively searched.

1. **The recorded uniform one-third band does not detect them.** Its stated constant is $\varepsilon_0 = \min\{1/4, c_{\mathrm{cal}}/1215\} \leqslant 1/4$, whereas $1/3 - h_j > 1/3 - h_0 = 27/100 > 1/4$. Thus setting $\mathscr{E} = 0$ does not contradict that inequality. In particular this is not another choice of topology inside its stated near-one-third strip.
2. **Both strict Hitchin–Thorpe inequalities hold.** $T_+/A_j \to 281/300 > 0$ and $T_-/A_j \to 961/2700 > 0$.
3. **The classical optimized scalar–Weyl moment bounds have slack.** Formula (14) gives $U_j > F(h_j)$. Choose a formal Einstein energy $e_j = (U_j + F(h_j))/2$; it has positive energies for both Weyl halves and satisfies the scalar and all functional-pencil nonnegativity bounds for large j. The equality case at $U = F$ was already classically forbidden; the new content is the uniform open strip $0 < U - F \lesssim h^2$. These moment values are a numerical non-implication witness, not a metric construction.
4. **The degree-to-complex-hyperbolic entropy obstruction has slack.** Collapsing the added summands gives a degree-one map $M_j \to X_j$. Kotschick's Corollary 3, equation (15), requires $\chi(M_j) > \frac{3}{2}|\tau(M_j)| + \frac{32}{81}\chi(X_j)$ for an Einstein metric. Here the left difference divided by A_j tends to $F_0/6 = 961/5400 > 32/243$, so this test is satisfied strictly. Entropy supplies a lower-volume restriction, not the missing upper bound used in (5).
5. **The elementary Seiberg–Witten Miyaoka–Yau test has slack.** $(\chi - 3\tau)(M_j)/A_j \to 59/1800 > 0$.

The mechanism is now different from a bare improvement of the $T_+/A = 2/3$ threshold: the actual class forces a quantitative lower energy envelope while the opposite Weyl half supplies a topological upper envelope. Their incompatibility eliminates every candidate Einstein metric on the same smooth type. It simultaneously rules out arbitrarily small normalized integral error on those types.

Sources and reproducible scope

- LeBrun, *Ricci Curvature, Minimal Volumes, and Seiberg–Witten Theory*, Proposition 3.2, Theorem 3.3 and Corollary 2.5. These provide the classical polarized gauge input, the one-third obstruction, and the exact equality comparison.

- Kotschick, *Entropies, volumes, and Einstein metrics*, Theorem 1 and Corollary 3. The comparison above uses its explicitly stated complex-hyperbolic degree bound, not an uncomputed exact minimal entropy for the new manifolds.
- Mal'cev's residual-finiteness theorem for finitely generated linear groups in characteristic zero supplies the finite covers; see Nica's account and proof. The unbounded-index deduction is given above.
- The nonclassical differential input remains the local functional-pencil proof, with its energy-envelope transfer. This note does not replace that proof by the finite calculations below.

`check_line1_two_chirality_topology.py` verifies the Chern–Weil and defect algebra, all rational ratios, the floor-error estimates, and the displayed comparison margins; it rejects incorrect opposite-orientation handle and trace-free-Ricci coefficients. It does not certify the analytic pencil theorem, Seiberg–Witten gluing or literature priority.